Learning Log

Year 1

Block B

[photo]

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«creativebrief/projectname»

Breda University of Applied Sciences

Project Brief:	Hyperlink
Assessment Assignment:	Hyperlink
Student GitHub Folder:	Hyperlink



How To Use This Template

As a student at Breda University of Applied Sciences studying to become a professional, you are required to provide evidence that demonstrates your professional learning and growth during the block. This template is intended to provide you with a well-structured and organised format for doing this effectively. Keep in mind that certain competencies may require you to update your evidence every week, and others may have only be relevant during a particular week or two.

With the evidence you present here teachers should get a clear and comprehensive overview of your progress, how you've engaged with feedback, as well as your general attitude and performance as a student and as an aspiring professional developer.

Note that number and size of text boxes and how they are organized on each slide may be modified as needed. It is up to you to decide what layout is most effective for the content you are providing. You may also add slides if needed, but try to be as economical as possible, i.e. quality over quantity. Focus on the things that are the most significant and meaningful.

You are also free to stylise this template to improve the graphic design and visual appeal, but please remember its purpose: to provide evidence of your progress towards becoming a professional developer. Changes should only enhance and extend the information, and may never subtract. I.e. do not delete anything or change the order of things. If in doubt, seek approval from your teachers!

Learning Log Structure

Section A
- Starting this block
- Goals

Section B
- ILO section
- Week log section

Section C
- Block reflection

Section A - My Plan *Must be completed in week 1*

Your plan describes your goals for the block. Starting with where you are right now, where do you want to be at the end of the block? I.e. what role(s) will you take responsibility for, and how will you demonstrate progress relevant to the ILOs for this block? What tasks and deliverables are best aligned with your role and the project brief?

Section B - ILO's *Must be completed in week 8, but should be updated regularly*

This is where you link your evidence to each of the Intended Learning Outcomes of this block.

Section C - Weekly Log *Must be updated every week*

This is simply a log (journal) of what you do each day, and what you learn from it. It is your responsibility to update this regularly and frequently. Be sure to include working links to any specific evidence you are referencing (and make sure that you share your docs with the teaching team). Note: some of the required information may be captured in production documents such as your work log or peer reviews.

In such cases, you simply need to provide links to those artifacts and may include any explanatory comment or reflection you feel is appropriate.

(Some reflection is almost always a good idea as it provides the foundation for Section C.)

Section D - Reflection *Must be completed in week 8*

Summative reflection on your progress during the block including a critical assessment of everything you did and learned during the block. This is a comprehensive review of everything recorded in Section B, evaluated against the goals and planning laid out in Section A.

In the first week, the student sets ambitious, S.M.A.R.T. goals in alignment with the project brief, their chosen role(s), content of the assessment rubric and their personal long-term goals

A

Section A

Goal Setting

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A Where am I starting?

Feedback from previous assessment -

Write on my learn log and work log daily and do not postpone my works for the last week.

My interpretation / key takeaways

Managing my time to step forward with data camp course.
Do not get behind of the content and fil work log and learning log every day and try to get feedback almost every session of data lab.

A What is my plan?

What I learned last block -

I learned how to work with Power Bi and I really liked that I know it is really useful in feature as a data analysis.
I practiced math and reviewing Python and I work in my self to how make a good presentation.

Previous block's grade 62,3

My current level of progress

In these two weeks off I also practiced Python and now I'm ready to retake a python exam and I feel comfortable now about this

Links

https://github.com/BredaUniversityADSAI/2024-25b-fai1-adsai-MohammadaliJaberi244437/blob/main/Graded%20Block%20A/244437_Mohammadali_Y1A_graded_rubrics_2024_2025.xlsx

A What are my goals?

S	**Specific:** Be specific when determining your goal. The more specific the better. Putting a real figure to your goal helps make it measurable. When specific and measurable your subconscious mind cannot override your goals, they become achievable.
M	**Measurable:** This where you define a measurable and meaningful goal. It's important to be able to track your progress. Without meaningful and measurable goals which you can track it's easy to lose heart and focus.
A	**Achievable:** Choosing realistic and short-term goals helps them be more achievable. Long-term goals can be larger and take more time to achieve. Planning smaller short-term goals as steps to larger long-term goals assists in achieving them.
R	**Relevant:** Each goal you set must be in line with your core values and in harmony with what YOU want out of life. Contradictions in values and goals leads to frustration and loss of motivation. Designing your goals with the expectation of your personal success.
T	**Time-Bound:** Setting a time limit on achieving a goal makes it measurable. When a goal is measurable you can SEE your progress and support your motivation and commitment. Giving your goal a time limit assists you in holding yourself accountable to its completion.
E	**Evaluate:** By evaluating your goals on a regular basis you will be much more likely to achieve them. It is easy to forget about long-term goals if they are not revisited from time to time. This also helps you assess if the goal you are working on remains in line with your core values and helps you measure your progress.
R	**Readjust:** Readjustments occur when goals are determined, through re-evaluation, to be misaligned with your values or long-term goals. This is especially helpful when you have been unable to achieve a goal in the time limits you set. This gives you the opportunity to modify your approach and increase your chances of success.

I am taking responsibility for *(if multiple add in percentages)*

Python Goal My goal for Python is to practice one to two hours daily to master it by the end of this block. While it might seem impossible to completely master Python due to the continuous challenges that arise, I will strive to improve and become proficient in it by the block's end. Based on my progress over the last two weeks, I would currently rate my Python skills at 60 out of 100. My goal is to reach 100 by the end of this block.

Another course we have in Block B is Machine Learning. I plan to review some individual Persian resources on this topic before the Data Camp course begins. This will help me feel more confident during the Data Lab sessions and allow extra time to practice. My goal is to complete at least one general Machine Learning course by the end of the first week. If I manage to finish one course during the Data Lab sessions, I will dedicate additional time to practicing. I also aim to spend at least one hour daily studying Machine Learning to become more comfortable with this subject.

Math is another critical subject in this block. After completing Machine Learning and Python, I plan to start a Khan Academy math course to strengthen my understanding. I will spend two hours daily working on math from Week 3 onward, in addition to my other studies, to improve in this area.

SQL is a subject I am somewhat familiar with, but I still need to practice and improve. Before the Data Camp SQL course begins, I will explore some Persian resources to deepen my understanding. I aim to practice SQL for one to two hours daily.

About making a deadline for my goals in Python and Machine learning I think I will be able to be Master in these two at the end of week 3 and Math and SQL because I found these two subject harder, I think I will be in a good position at the end of block B

Personal and professional development goals

I mention my personal and professional goals above.

B

Section B

Progress Log

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1 Week 1 - Log

What did you do?

- What goal(s) did you set for this week?

For this week I hope I can watch courses about Machine larning and try practicing Python daily.

- What have you actually been able to do?

I was able to do Python but I wanted to work more on it to be more conident in this major.

- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

1 Week 1 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response to Feedback

Click for feedback delivery techniques!

2 Week 2 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn't go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

2 Week 2 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

3 Week 3 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

3 Week 3 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

4 Week 4 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

4 Week 4 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

5 Week 5 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

5 Week 5 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

6 Week 6 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

6 Week 6 - Feedback

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

7 Week 7 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

8 Week 8 - Log

What did you do?

- What goal(s) did you set for this week?
- What have you actually been able to do?
- Showcase the evidence of your progress (production artifacts, short descriptions-links-pictures animated gifs, etc.)

Reflection

- How did the week go?
- What went well?
- What didn’t go so well?
- What did you learn?
- What could be added as an Action point looking forward to next week?

Team, individual, peer, workshop feedback

- **[Reviewer]** Feedback

Response

Click for feedback delivery techniques!

C

Section C

Progress
Intended Learning Outcomes

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0

ILO 0.2

Foundational skills

The student masters the foundational skills and knowledge that form the basis of Data Science and Artificial Intelligence.

ILO 0 Foundational skills *The student masters the foundational skills and knowledge that form the basis of Data Science and Artificial Intelligence.* 1/1

0.2

The students is able to demonstrate an advanced understanding of python programming concepts and basics of SQL.

pre-requisites	A (Points: 3)	B (Points: 3)	C (Points: 3)	D (Points: 3)	E (Points: 3)
The student should submit the relevant Python Notebooks addressing the creative brief. Student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	The student is able to demonstrate an understanding of the concept of working directory, file types and file structures in Python.	The student is able to perform basic data cleansing activities in Python with or without using packages. Additionally, the student is able to demonstrate significant understanding of Numpy.	The student is able to design reports and visuals (e.g., using matplotlib, Streamlit) based on the merged datasets which help providing valuable and actionable insights.	The student demonstrates the capability to establish a connection between Python and a database, perform basic ETL operations such as read/write.	The student is able to join two different tables using SQL and uses this to fulfill the requirements of creative brief.
Justification +Links	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**

1

ILO 1.2

Professional practice

The student can collaborate (internationally) in multidisciplinary teams with different levels of knowledge in the field of data use and applications. They can set up and execute projects in collaboration with stakeholders and team members. They can act as a sounding board in discussions with team members, customers, users and experts. They strive for a good balance between input of their own vision and additional expertise of others. They are able to lead a team.

ILO 1	Professional practice	*The student can collaborate (internationally) in multidisciplinary teams with different levels of knowledge in the field of data use and applications. They can set up and execute projects in collaboration with stakeholders and team members. They can act as a sounding board in discussions with team members, customers, users and experts. They strive for a good balance between input of their own vision and additional expertise of others. They are able to lead a team.*	1/1

1.2 The student accurately tracks time spent on planned and unplanned work, noting reasons for and consequences of deviations from the planned objectives.

pre-requisites	A (Points: 2)	B (Points: 2)	B (Points: 2)	B (Points: 2)	B (Points: 2)
The student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	The student outlines the completed tasks in the worklog, with detailed comments provided where necessary.	The student writes professional and critical reflections with time-tracking and identifying key lessons.	The student fills out the worklog for each of the weeks.	The student includes all references to important resources in the worklog for the listed tasks.	The student's writing style in worklog is professional and free of spelling and grammar mistakes. The student comprehends what was completed and why individual and project feedback was given.
Justification +Links	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**

2

ILO 2.2

Personal development & Academic practice

The student applies relevant (research) methods and techniques in combination with relevant and adequate argumentation. They can reflect on (business) processes and their role in them, both theoretically and practically, by constantly evaluating their own actions and adapting them with input from others. They can translate the result of the reflection into concrete personal learning objectives.

ILO 2 Personal development

The student applies relevant (research) methods and techniques in combination with relevant and adequate argumentation. They can reflect on (business) processes and their role in them, both theoretically and practically, by constantly evaluating their own actions and adapting them with input from others. They can translate the result of the reflection into concrete personal learning objectives.

2.2

The student shows knowledge of leadership skills by taking on the role of a leader and supporting others in being a leader.

pre-requisites	A (Points: 5)	B (Points: 5)
The student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	Goals written with a **SMARTER** method. The goals are shown to be Specific, Measurable, Achievable, Relevant, Time-bound, Evaluated and Readjusted. Project goals and self-development goals detailed.	Goals are aligned with project, personal, and professional development objectives. A detailed evaluation of the goals has been conducted throughout the block. The goals may have been re-adjusted as well.
Justification +Links	**Justification +Links**	**Justification +Links**

6

ILO 6.2

Data Collection and Processing

The student masters (technological) skills to acquire, pre-process, process and manage the necessary data to create value for individuals, organizations and domains.

6.2

The students is able to perform basic ETL operations such as read/write, clean, and explore data using standard Python libraries.

pre-requisites	A (Points: 5)	B (Points: 5)
The student should submit the relevant Python Notebooks addressing the creative brief. Student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	The student is able to clean and pre-process and visualise data using Python libraries.	The student is able to use metadata from the dataset to analyse the different types of data with describing the role (e.g. id / feature/ target) as well as detecting missed values.
Justification +Links	**Justification +Links**	**Justification +Links**

8

ILO 8.1

Modelling

The student can apply modelling techniques including Machine Learning and AI to create value for individuals, organizations and domains.

ILO 8 Modelling

The student can apply modelling techniques including Machine Learning and AI to create value for individuals, organizations and domains.

8.1

The student is able to understand different types of supervised and unsupervised machine learning algorithms for data analysis and can implement these to solve a business objective using industry standard Python libraries.

pre-requisites	A (Points: 4)	A (Points: 4)	A (Points: 4)	A (Points: 4)	A (Points: 4)
The student has a sufficient knowledge about ML algorithms. The student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab. .	The student is able to understand regression concepts and build a regression baseline using dataset.	The student is able to understand classification concepts and build a classification baseline using dataset	The student is able to understand clustering concepts and build a clustering baseline to segment patients into clusters using dataset.	The student is able to do the feature selection and engineering in dataset and is able to improve their model by tuning its hyperparameter(s) using a validation set and justify their model improvement(s) in the context of the business objective.	The student is able to analyse and interpret the results obtained from the ML algorithms for model selection based on the solution requirements and dataset features.
Justification +Links	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**

8

ILO 8.2

Modelling

The student can apply modelling techniques including Machine Learning and AI to create value for individuals, organizations and domains.

ILO 8 Modelling

The student can apply modelling techniques including Machine Learning and AI to create value for individuals, organizations and domains.

8.2

The student is able to perform elementary operations on matrices and understand the concepts of derivatives, gradients, optimization algorithms, and can implement a simple machine learning model from scratch.

pre-requisites	A (Points: 4)	A (Points: 4)	A (Points: 4)	A (Points: 4)	A (Points: 4)
The student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	The student demonstrate a comprehensive understanding of key optimization algorithms, critically analyse their performance, and apply them to solve complex minimization and maximization problems. Furthermore, the student will have the capability to assess the impact of algorithm parameters on performance and convergence in mathematical optimization.	The student implement all steps of training a regression model from scratch for their project, including implementing the cost function and finding its derivative, training the model using an optimization algorithm, evaluating the model's performance using appropriate regression metrics, and performing accurate predictions based on the trained model.	The student conduct fine-tuning of hyperparameters for regression models implemented from scratch, which includes implementing the regularized form of the cost function, testing different optimization algorithms (modifying them as necessary when incorporating regularization), studying the impact of learning rates on training performance, and examining how scaling features influences model performance and generalization.	The student implement all steps of training a classification model from scratch for their project, including defining the cost function and calculating its derivative, training the model using an optimization algorithm, evaluating the model's performance with relevant classification metrics, and making accurate predictions based on the trained model.	The student carry out hyperparameter fine-tuning for classification models developed from scratch, focusing on testing various optimization algorithms (modifying them as necessary to integrate regularization), implementing the regularized form of the cost function, analyzing the effects of learning rates on training performance, and assessing how scaling features affects model performance and generalization.
Justification +Links	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**

11

ILO 11.1

Reporting and Advising

The student can translate (interim) results into effective reporting. They see opportunities and possibilities and can translate these from a market-oriented vision into new concepts, products or services, while keeping the business side of the organization in mind.

ILO 11 Reporting and Advising

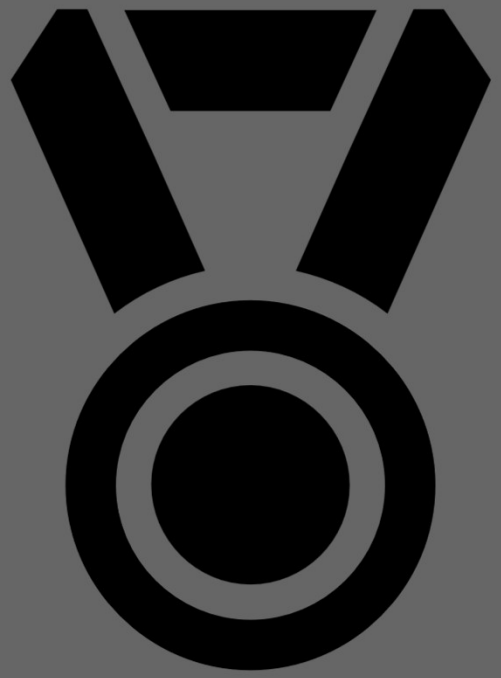
The student can translate (interim) results into effective reporting. They see opportunities and possibilities and can translate these from a market-oriented vision into new concepts, products or services, while keeping the business side of the organization in mind.

1/1

11.1

The student is able to communicate technical concepts by means of a structured, coherent, and well-formatted professional report.

pre-requisites	A (Points: 3)	B (Points: 3)	C (Points: 3)	D (Points: 3)	E (Points: 3)
The student should present final project and be able to answer the questions related to it during the presentation. The student submits a complete learning log, work log and self-assessment rubric. The student exhibits professional behavior during the DataLab.	The student is able to define the sections (EDA, Regression, Classification, Clustering, Evaluation, Improvement) clearly based on implementations.	The students is able to demonstrate an advanced understanding of machine learning concepts.	The student is able to demonstrate understanding of different types of machine learning algorithms for data analysis and have implementation of baselines based on the defined project and dataset.	The student is able to critically analyse and evaluate their proposed solutions based on its relevance for the business objectives. .	The student has a clean and commented code and is able to identify the shortcomings of the current implementation and propose next steps for improving the quality of implementation.
Justification +Links	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**	**Justification +Links**



Medal Challenges

Going the extra mile!

Put your evidence down here to receive a medal!

Show your best examples, do not go overboard, add in short description, you are free to alter this layout (or add slides per evidence) to suit your needs. Just be sure that it is clear.

Section D

Reflection

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D What did I achieve this block?

My most significant achievements this block

...

My most difficult challenges this block

...

The most important lessons I learned

...

D How well am I progressing?

Reflection on my self assessment

...

How I plan to improve next block

...

My self assessment grade is a

X.X